

# LLM

# Function calling

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# Agenda

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- 为什么学 LLM function call -Agent
- 一个最简单的llm function call
- 简单的LLM function call 工程
- LLm function call 进阶架构 - React Agent
- LLm function call 进阶架构 - Plan and execute
- LLm function call 进阶架构 - refelection
- LLm function call 进阶架构 - 多角色轮询

# 为什么学 LLM function call -Agent

'21      '23      '24    ... 2026

\_\_\_\_\_ LLMs

\_\_\_\_\_ Compound AI sys.

\_\_\_\_\_ Agents

A Generative AI Agent is defined as an **application** that tries to achieve a **goal** by **observing** the world and **acting** upon it using the **tools** it has at its disposal.



# Agent types

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## Modality Based

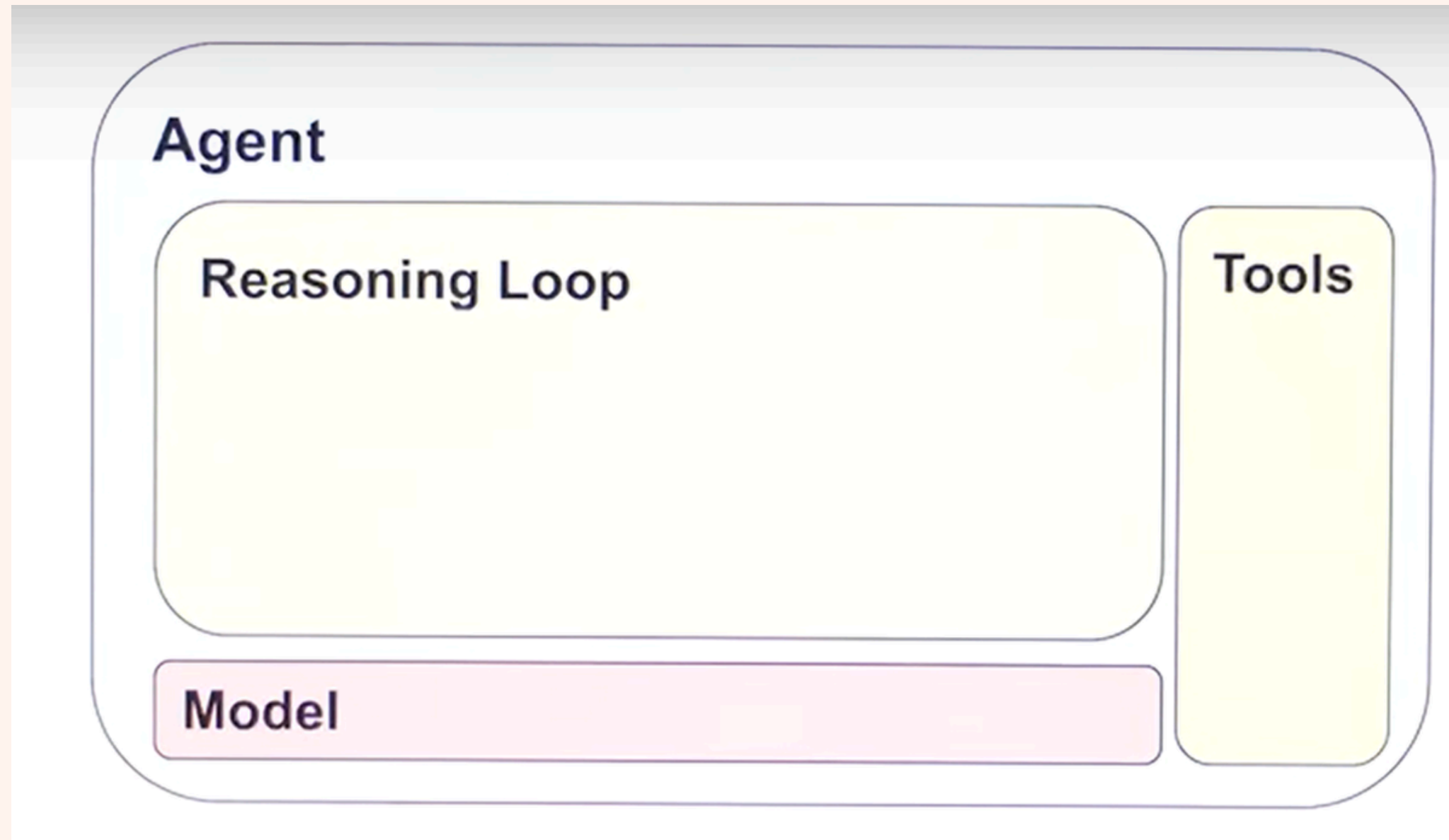
- **LLM (Large Language Model)**, focus on language inputs
- **VLM (Vision Language Model)**, can take both Vision and language inputs
- **VLA (Vision Language Action)**, combine vision and language to produce physical actions, often applied in robotics

## Use Case Based

- **Gaming**, NPCs in games
- **Healthcare**, Diagnostic agents
- **Manufacturing**, VLA agents for assembly lines
- **Cybersecurity**, Adversarial testing, red teaming
- **Customer Support**, LLM-based agents
- **Software Development**, multi-agent frameworks for writing code

# Agent components

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# Model types

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Any Generative Language model (large or small)

- gpt, Claude, Gemini, LLaMA can vary in size and capabilities.

Models can be multimodal or finetuned:

- audio to text; text to audio – hugging face

Training data can play a large role

- Models trained on datasets that include tool usage and reasoning loops often exhibit enhanced reasoning and problem-solving abilities.



# Tools

## Extension (扩展)

- 增强AI能力
- 复杂、持续性任务
- 可能需要外部资源
- 例如：网页浏览、数据分析

## Function (函数)

- 执行特定任务
- 简单、离散性操作
- 通常是内部处理
- 例如：计算、格式转换

## Datastore

- 存储私有数据
- 高效检索
- 增强理解
- 动态更新
- 集成与分析



Drive



Gmail



Search



Cloud  
Functions



Cloud  
Run



Code  
Interpreter



JSON



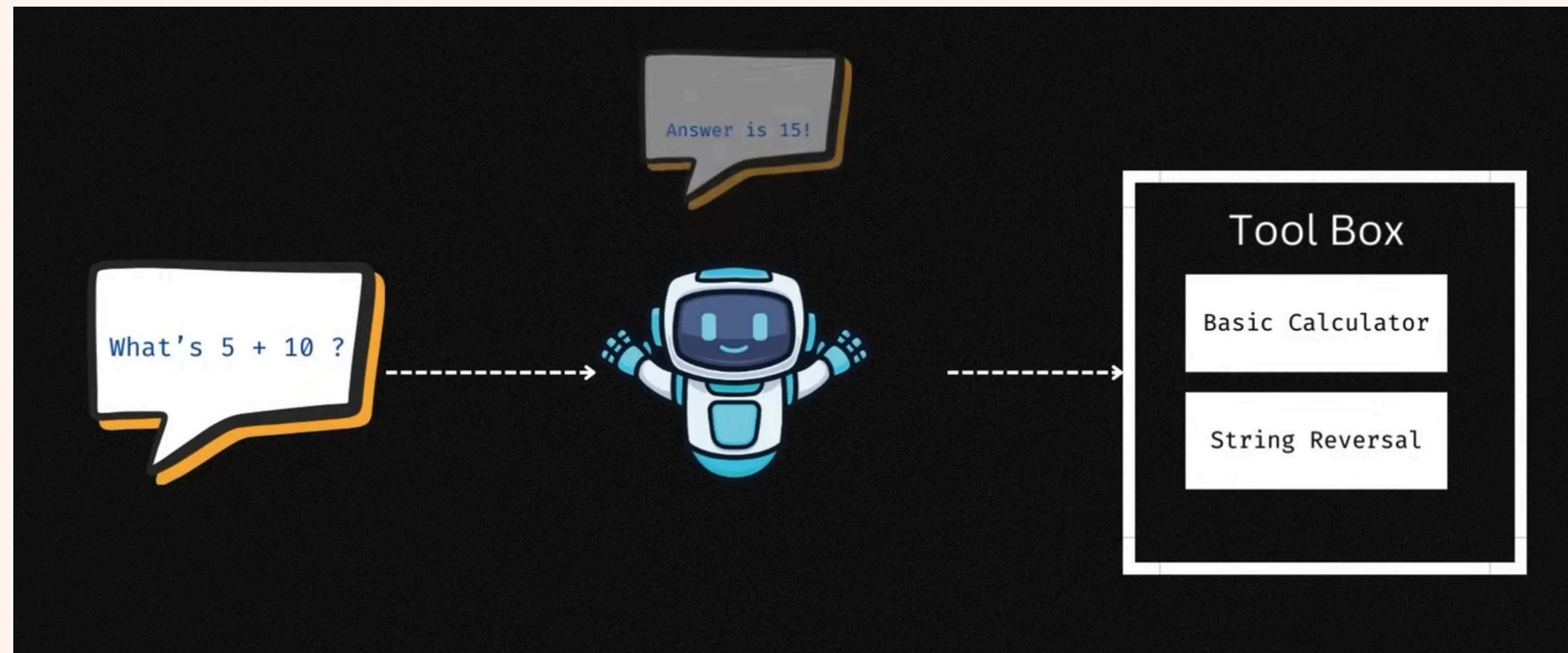
Vertex AI  
Data Stores

# 一个典型的例子

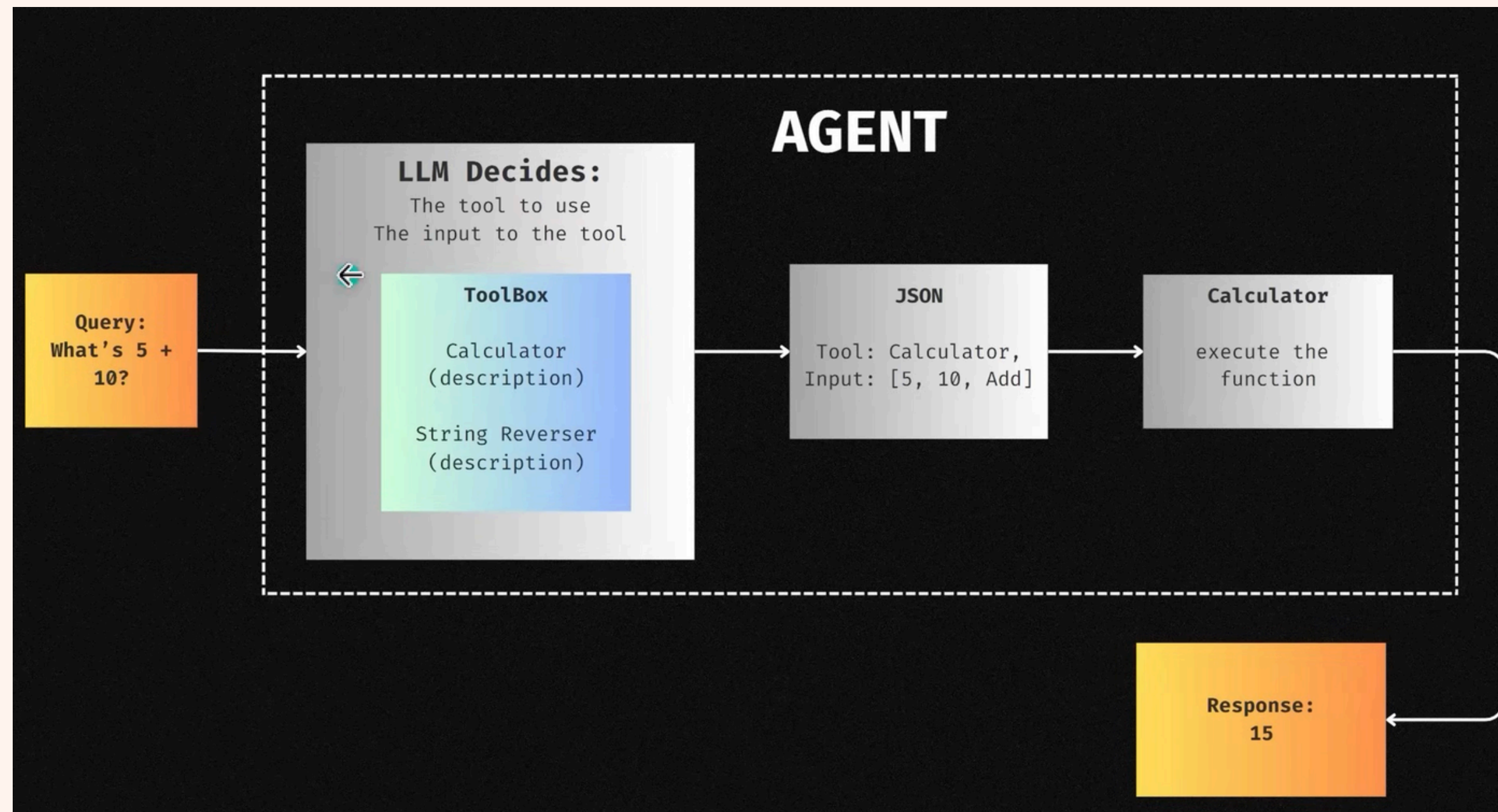


# 一个简单的 LLM function call - Agent

# A Simple AI Agent with 2 function calls







# 简单的LLM function call 工程

# 一个简单的 LLM function call - Agent

# LLm function call 进阶架构

## - React Agent



# LLm function call 进阶架构

## - Plan and execute

# LLm function call 进阶架构 - refelection

# LLm function call 进阶架构

## - 多角色轮询

# Thanks